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Real-time plasma diagnostic tool for the investigation of azimuthal rotating instabilities in Hall Effect thrusters

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Abstract

The following paper reports about a non-intrusive diagnostic system able to measure the frequency of azimuthal instabilities inside the discharge channel of Hall Effect Thrusters (HETs) up to several hundreds of kHz in a real-time fashion. The innovative optical assembly, formed by an optimized sensor array, can help to drastically reduce the amount of processed data and the size of the opto-electronic front-end. The performances of the diagnostic system have been tested successfully during several tests on a HET in different regimes, showing an accuracy and reliability comparable with the state-of-the-art video recording methods. In order to test the real-time fashion of the diagnostic, we have monitor the time-evolution of the azimuthal mode spectra during the startup transient of the thruster, highlighting a redistribution of the oscillation energy among the different mode to an asymptotic stationary regime. The high compactness of the developed sensor array opens new possibilities to perform simultaneously multi-channel or species-selective real-time measurements, while the reduced electronics and power consumption pave the way to miniaturized tools for on-flight diagnostics.

Keywords: Thruster, Plasma, Electric propulsion, Hall effect, Diagnostics, Non-intrusive, Azimuthal instabilities

Introduction

Non-intrusive diagnostic tools for ion thruster engine are interesting since they allow reliable and high fidelity analysis of thruster's performance and a deeper understanding of Hall Effect thruster (HET) internal dynamics, without altering the ejected plume [1–3]. The most popular non-invasive diagnostic tools are laser induced fluorescence (LIF) [2, 4, 5], spectral-imaging [6, 7] or video-based systems [8, 9].

Spatially resolved spectral methods offer the most complete diagnostic solution, providing a full 3D, high-resolution reconstruction of the plume parameters (ion-velocity vector map, electron temperature and density), both inside the discharge channel and in the near-field region. A spatial averaged value of the internal parameters of the plume can be also extracted from the relative intensity of the spectral lines of the radiation

emanating from the thruster, with the advantage of a way simpler experimental setup but at the cost of the spatial resolution [10].

Azimuthal rotating instabilities in HETs

In the discharge channel of HETs a large plethora of observable phenomena is present. The plasma dynamics is often characterized by azimuthal oscillations, also known as “azimuthal rotating spokes”, occurring at frequencies larger than kHz. In particular, a bifurcation of the azimuthal instabilities into low- and high-frequency branches has been observed in this type of thrusters by several research groups in the world. The low-frequency branch, characterized by rotating spokes with frequencies ranging from ten to hundreds of kHz, has been deeply investigated in the recent years by researchers at the University of Michigan [11, 12] and at the French National Centre for Scientific Research [13]. A parallel line, begun by Adam et al. at the Ecole Polytechnique [14, 15], are instead investigating azimuthal waves at megahertz frequencies, exploiting kinetic models of Hall thrusters and performing experimental observation of the same instabilities using collective light scattering techniques. In general, the characteristics (oscillation frequency and amplitude) of rotating instabilities are strictly dependent on the thruster operating parameters, i.e., discharge voltage, magnetic-field strength (or magnet-coils current), anode and cathode mass flow rates. Vice-versa, electron dynamics in the Hall thruster plume could affect every aspect of thruster operation [11]: the ease of electron transport across magnetic field sets equi-potential lines that guide ion acceleration and focusing, determining beam divergence and affecting channel erosion patterns.

The mechanism for spokes formation, however, is still debated since azimuthal and axial modes result from a complex interplay of various phenomena and an understanding of these complex interactions remains far from complete [16–19]. In previous works [11, 20], two types of mechanisms are generally proposed to explain anomalous cross-field electron transport in Hall thrusters: collisional transport via electron impact with thruster surfaces and turbulent transport via fluctuating azimuthal electric fields. Turbulent transport is a collision-less transport mechanism where induced fluctuations in the electromagnetic fields of the Hall thruster plasma create axial $E \times B$ drift components, carrying electrons across the applied magnetic field to the anode. Recently, some researchers at the University of Michigan are trying to develop novel first-principles models of the anomalous electron transport, testing their validity via numerical simulations [21]. In this scenario, the development of improved systems for the diagnosis of azimuthal instabilities could provide a great help not only to understand clearer the physics behind the HET principles, but also to refine the HET design.

Observation of azimuthal rotating spokes

In several previous works, pioneered by the group of researchers from the University of Michigan [13, 20, 22], the investigation of rotating spokes in discharges channel of the HETs was made possible by using video cameras featuring extremely high frame-rates, which are able to capture the time evolution of light emission inside the HET's channel. The light intensity profile emanated from the thruster, generated mostly by collisional phenomena, can be related to the ion density distribution inside the discharge channel. Moreover, a nice feature of the video approach is that the recording system can be

placed outside the vacuum chambers avoiding all the additional complications related to the development of vacuum-compatible hardware and electronics.

Although the ultra-high-speed imaging system is now a consolidated technique, providing valuable information about the dynamics of specific thrusters, very high frame rate acquisitions correspond to extremely large datasets, whose processing requires a significant computational effort. In order to overcome this issue, in this work we propose a different scheme, where the collection of the light emitted from the HET discharge channel is performed by a custom optical assembly and a dedicated sensors array. The use of a sensors array with a smaller number of sampling points, as discussed in a previous work [2], leads to a significant reduction of the amount of post-processed data and it makes possible to reach easily very high sampling rate, while maintaining a real-time insights of the dynamics during the thruster operation. The performances of the developed sensor, in terms of sampling frequency and refresh rate, have been tested in two different types of measurements. In a first session of measurements, we have measured rotating spokes spectra at different operating points of the thruster in a stationary regime. This type of tests are useful to evaluate the quality of the obtained results, estimating the signal-to-noise ratio and comparing the observed spectra with previous works that are present in literature. In a second session, we have instead monitored the time-evolution of the azimuthal mode spectra during the startup transient of the thruster. The reported results represent a better insight of the novel approach allowed by a real-time observation. As an example, we report in [Real-time measurements on a 5 kW HET in different operating regimes](#) section about the evolution of the energy and the rotating velocity in each mode over time. Such type of information could be useful to better understand mode-hopping behaviors that can occur in the thruster or to perform a real-time optimization of the operation.

Finally, combining the high performances with the compactness of the optical assembly, multiple sensors can be exploited to perform different type of measurements: for example, using wavelength selecting filters and many array of sensors, it is possible to monitor azimuthal oscillations of different types of particles (different atomic species or different ionization levels) or more discharge channels simultaneously. Additionally, the presence of a reduced hardware opens to the possibility to develop diagnostic tools suitable for a direct-in-orbit monitoring of plasma rotating oscillations.

Experimental apparatus for the rotating spokes diagnosis

In the McDonald technique [20] for high-speed image analysis, in each frame of the video the profile of the discharge channel is isolated and an additional radial integration stage is performed, in order to reduce the size of the data. However, the use of pre-processing procedures, required to discard the unnecessary bits that do not belong to the investigated spatial region, may result in slow and resource-demanding tasks, limiting the possibility to implement a real-time analysis, that is instead the goal of this work. Commercial of the shelf camera, bought for example by iX Cameras or Phantom, usually can achieve a maximum frame rates of the order of several hundreds of kSPs/s with an image size of 0.1 MPixels, correspondent to a memory allocation of 125 kbyte. Therefore, if we consider a frame-rate of 400K frame-per-second, the streaming of the data should sustain a speed larger than 100 Gb/s in order to provide a real-time acquisition.

Moreover, the large size and weight of such type of cameras, usually of the order of kilograms, and their power consumption, that exceeds easily 200–300 W, cannot fit the requirements for direct-in-space diagnostic applications.

Optical front-end

The optical front-end developed in this paper replaces the CMOS sensor with an array of discrete optical sensors, that can be formed by a combination of multi-mode fibers and a photo-multiplier (PMT) or directly by an array of high-speed photo-diodes. Rather than collecting an image of the entire HET system (i.e., the discharge channel and its surroundings), the developed system samples directly the discharge channel region at discrete points that are uniformly spaced along the channel (Fig. 1a). Obviously, the detail of the ion density spatial distribution is no longer visible but, since the goal of the presented diagnostics is the measurement of amplitude and the frequency of the rotating spokes, the loss of resolution does not affect the Fourier analysis.

The optical front-end exploited in this work is represented by an array of eight silicon photo-diodes (Fig. 1b), that are optimally designed without the metallic housing and whose active area is chosen to provide the best compromise between responsivity and low junction capacitance. In literature [10–13, 16], high-power HETs usually exhibit azimuthal currents that can be described in terms of harmonic modes with orders rarely exceeding the 4th order: it can be shown that eight sampling points uniformly spaced across the discharge channel allow to measure harmonic modes up to this order.

The use of the photo-diodes array instead of the bundle of fibers [2] significantly increases the amount of fluorescence light collected by each sensor, due to the larger active area, and it allows to improve the compactness of the whole electronic platform (no avalanche photo-diodes (APD) or PMT are needed).

Since the light emitted by the thruster discharge channel is collected through a viewport placed at an angle of 45° with respect to the fire axis, the sensors array features an elliptical geometry (Fig. 1b), that matches the image of the discharge channel obtained by the photographic objective. In particular, the exploited photographic objective is a telephoto lens, featuring a magnification of 0.1 and a focal length of about 300 mm. It has been developed with telecentric properties on the image plane, in order to project the image of the thruster on a plane orthogonal to the optical axis, even if an angle between the object and the viewer is present. On this plane, the sensors are aligned by using a two-axis micro-metric translational stage, looking for the position that maximizes the signal acquired from each photo-diode. With the current magnification, each sensor measures over a spatial area of 5% of the channel, as represented in Fig. 1c).

However, even with the telecentric correction, the presence of the large angle of view can cause a non-homogeneous illumination of the sensors, leading to possible artifacts in the measured spectra, as explained in [Real time fourier analysis](#) section. Indeed, since a large quantity of the fluorescence is emitted some millimeters behind the exit plane of the channel, few photo-diodes collect a larger amount of light from the inner part of the channel, while others are illuminated only by a more superficial emission (see Fig. 1c)). As explained in [23], the relative fractional distribution of ions and atoms depends on the distance from the exit plane. In addition, considering that in all the cases analyzed in this work the electron temperature is below 20 eV, their relative contribution in the

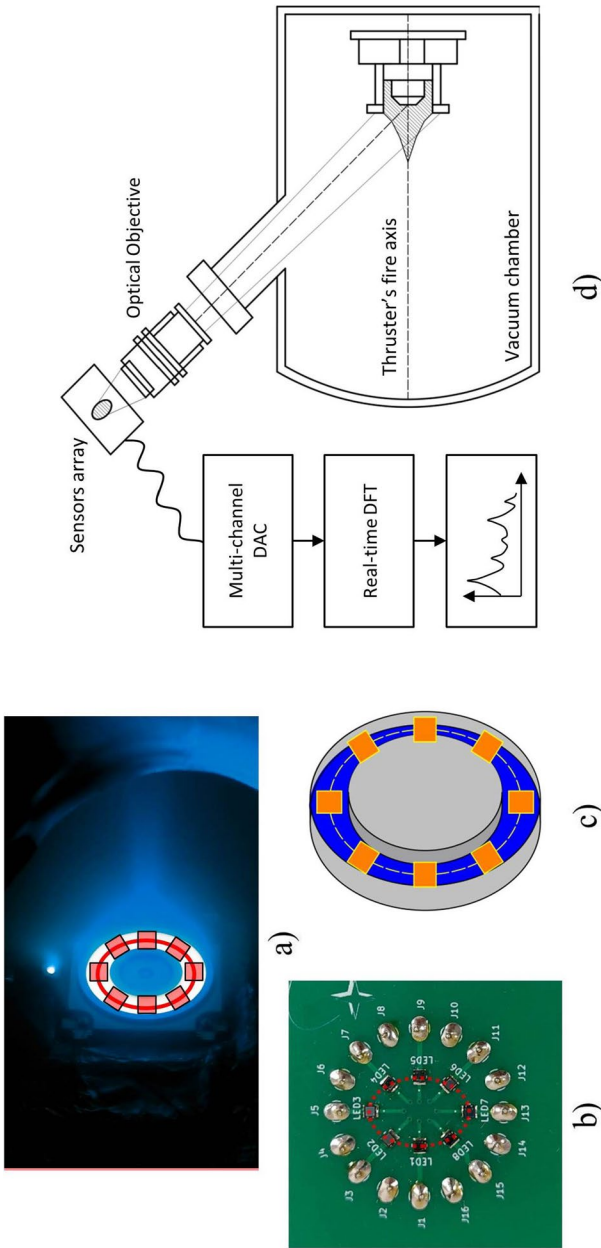


Fig. 1 **a** Picture of a HET in the vacuum chamber with the position of the sensors array overlapped, explaining the concept of the discrete sampling; **b** PCB with the eight photo-sensors; **c** Realistic drawing of the superposition between the sensors array and the image of the HET taken with a 45° angle of view; **d** Scheme of the entire experimental setup for the real-time analysis of the azimuthal rotating spokes

emitted light intensity depends on the local electron temperature. Therefore, since the developed optical sensors have a large sensitivity both in the visible and near-IR region, the diagnostic system is sensible to the radiation emanated by the relaxation of both the ions and neutral particles. However, we have estimated that our optical assembly collects the emitted radiation up to 5 mm inside the exit channel: at this depth [23], the contribution in the light intensity provided by the ions is much larger than the one provided by the neutrals. We can therefore assume that the optical assembly is measuring mainly the azimuthal motion of the ions inside the channel. Nevertheless, due to the large integration area and spectral bandwidth of the sensors, the measurement of the azimuthal frequencies could be slightly influenced by the radiative contribution of the neutral axial distribution's tails: this additional effect could be one of the source of the impurities that are present in the observed spectra reported in the next sections.

Finally, the electrical signals generated by the photo-diodes is then amplified with a dedicated electronics and then they are sent to a high sampling-rate (up to 800k kSample/s) eight-channel simultaneous-sampling analog-to-digital converter (ADC). Finally, the digitalized signal is transferred to a dedicated computer that processes it for a real time Fourier analysis (Fig. 1d)), explained in the [Real time fourier analysis](#) section.

Real time fourier analysis

The real-time Fourier processing consists in a highly simplified version of the analysis proposed in Ref. [20]. For each acquisition, the light intensity measured by a single photo-diode generates a vector $V_j(t_i)$ (Fig. 2a), where $j = 1...8$ corresponds to the sensor number and $i = 1...N$ labels the time vector. N is the number of the acquired samples that depends on the frame-rate and the time window chosen for the acquisition. In all the reported measurements, we have chosen time-windows of tens of milliseconds, that allow to measure the spectral density with a resolution of fraction of kHz.

After the digitization, the DC-component is first removed from the temporal traces that then undergo a normalization. The procedure generates a set of traces $\tilde{V}_j(t_i) = (V_j(t_i) - \langle V_j \rangle) / \max(V_{ij})$, where $\max(V_{ij})$ is calculated over the entire set. After those passages, it is possible to directly reconstruct the “Spoke-Map” $\tilde{V}_{ij}$, forming a 2D matrix where each row corresponds to the AC-values of the sensors at a certain instant, while the column represents the intensity of the collected light by the j th photo-diode as a function of time, in analogy with video-based processing techniques [11, 20]. A 2D Fourier transformation is finally applied to the matrix $\tilde{V}_{ij}$. The 2D Fourier decomposition separates all the oscillating modes and it produces a matrix with the same dimension of $\tilde{V}_{ij}$, where each column corresponds to the Fourier amplitude as a function of the frequency of a single spatial mode.

An example of the real part of the 2D-DFT matrix, measured during a test performed in a Aerospazio Tecnologie Vacuum Test Facility, is shown in Fig. 2b. The DFT matrix is symmetric and redundant across its center, thus only the unique first quadrant is reported. The selected data has been measured during the optimization sequence of the operating parameters of the thruster, where it has exhibited a lot of oscillations. The modulus of 2D DFT provides a very quick method of reading the results, where the vertical axis indicates the frequencies and the horizontal one indicates the order of the azimuthal mode. As can be seen, the contrast achieved by our

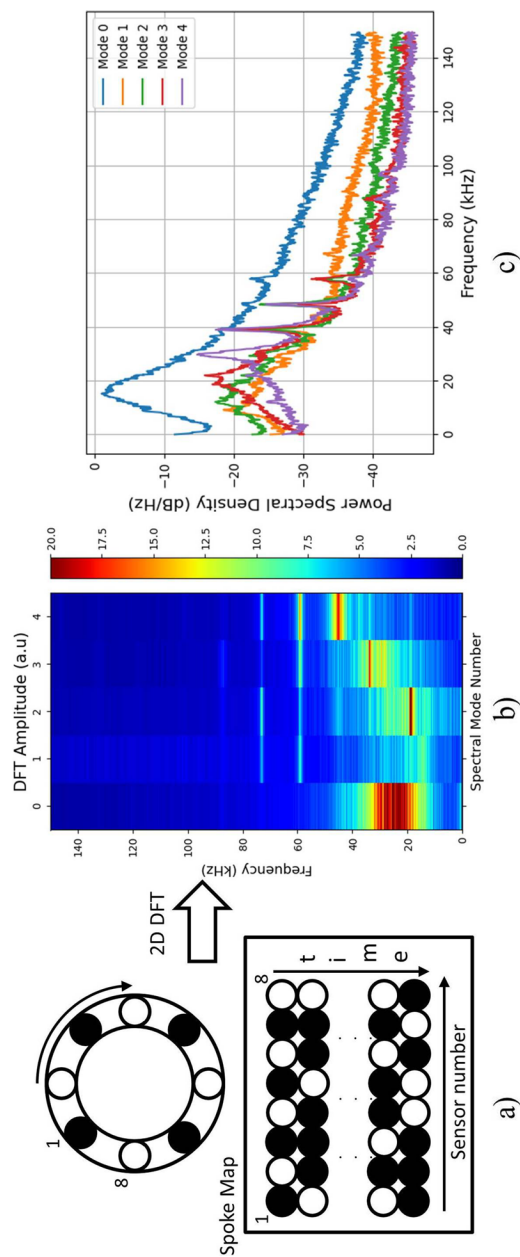


Fig. 2 **a** Conceptual explanation of the spoke map created with a discrete sampling acquisition. Example of the modulus of the 2D DFT in **(b)** and the correspondent Power Spectral Density of the first azimuthal rotating modes in **(c)** acquired with the developed diagnostic system

system is at the same level that the one obtained by the high-speed video recording and it allows to characterize the results with a nice signal-to-noise ratio.

The quality of the results is highlighted also looking at Fig. 2c, where we report the Power Spectral Density (PSD) of each individual mode. In this case, the signal to noise ratio is increased taking an average on about thirty acquisitions and applying a low pass filter. The spectra of higher modes show very clearly the presence of rotating spokes at well-defined frequencies. However, it is possible to observe that some peaks with the same frequency can be found in spectra belonging to different modes. This effect can be mainly due to two different causes: the first one is the presence of some cross-talk (< -20 dB) generated by slightly different electrical responsibilities of the photo-diodes or due to the non-perfectly homogeneous illumination of the sensors, for the reasons explained in [Optical front-end](#) section. The second one can be instead attributed to the high complexity of the dynamics inside the discharge channel, where modes with different wave numbers are not completely independent each other. This second supposition is also supported by the fact that such cross-talk has been also observed in previous experiments performed with video-based systems where the thruster has been imaged orthogonally respect to the fire axis and the data have undergone a detailed post-processing [10–12].

Regarding the processing speed, the bandwidth of the amplified photo-diodes (of the order of 1 MHz) and the sampling-rate of the ADC allow to measure power spectra up to 400 kHz, that is a large enough frequency for the type of spokes investigated in this work, with a single-shot refresh rate of the order of few Hz. After the aforementioned averaging procedure over many acquisition, the complete analysis provides one averaged spectrum every few seconds. Even if the achieved refresh rate can be seen as a limitation to the real-time fashion of the developed technique, we would like to emphasize that the resulting performances are strictly related to the current hardware used for the demonstration. The main limitation is indeed represented by the time that is needed to transfer the acquired data from the digitizer to the computer and the time required to perform the Fourier Analysis, that depends on its calculation speed. The optimization of the mode decomposition method [24] and the pairing of the optical assembly with a real-time processor (i.e. an FPGA) could help to speed up the process quite easily.

In conclusion, the results show that the real-time fashion of the high-speed diagnostic system allows to characterize the oscillating plasma dynamics occurring in the thruster discharge channel in either stationary or transient operating condition, as demonstrated in the next chapter.

Real-time measurements on a 5 kW HET in different operating regimes

In order to validate the reliability of the new diagnostic system, it has been exploited during multiple tests, where HETs were operating at several operating points, covering different values of discharge voltage V_d and current I_d . The diagnostics tool has managed to provide a almost real-time insight of the oscillating dynamics, allowing to monitor constantly the rotational instabilities over many hours (or days) of operations.

Rotating spokes spectra at different operating points

In a first session of measurements, the real-time diagnostic has been used to perform a complete characterization of the oscillation dynamics at different operating points. Within the large amount of collected spectra, three main different oscillating behaviors have been observed during the test, reported in Fig. 3, where we show some examples of the measured spectra.

The first type of spectra (Fig. 3a) is characterized by the presence of a very shallow global oscillation (correspondent to the “Mode 0” or “breathing” mode) and the absolute absence of higher harmonic frequencies. This behaviour has been more frequently observed when the thruster is operating at low discharge voltage and high current.

When the discharge voltage is increased, the power spectral density starts to become populated by narrower frequency components. In particular, the second type of spectra that we have observed (Fig. 3b) is formed by very high and sharp peaks associated to the breathing mode. The main breathing mode frequency and the relative higher harmonics are clearly visible up to 100 kHz. In this condition, the whole amount of energy remains in the breathing mode and all the higher modes are suppressed. It can be noted that the strong breathing mode signature is visible also onto the higher modes: this can be due to the cross-talk explained in the [Real time fourier analysis](#) section, that in this regime becomes more visible.

Finally, the most interesting condition is shown in (Fig. 3c) where, in addition to the breathing mode, we observe a clear signature of several peaks in the higher modes spectra, highlighting the presence of azimuthal rotating spokes. The amplitude of these peaks is quite smaller respect to the breathing mode but they appears at well-defined frequencies, which are clearly not originated by an eventual cross-talk.

Collecting data from different regimes, it is possible to retrieve some common trends:

- The appearance of rotational azimuthal instabilities is more probable when the thruster operates at high discharge voltage and at low current;
- The high-current condition is the closest to a no-spokes condition;
- Higher spokes modes rotates at higher frequencies.

The observed trends are in agreement with the experimental results reported in literature [11–13] for HETs with different geometries, validating the reliability of the developed diagnostics.

Time evolution of the spokes during the transient

In order to test the real-time features of the developed diagnostics, we observed how the instabilities of the plasma evolve at the switching on of the thruster. An other possible application could be, for example, the evolution of the azimuthal instabilities during a passage between two different operating points. However, such type of measurements is not possible in the tests examined in this work, since they follow a standardized procedure: once the thruster has reached the stationary regime, it is switched off in order to permit the measurement of the thrust by a differential method.

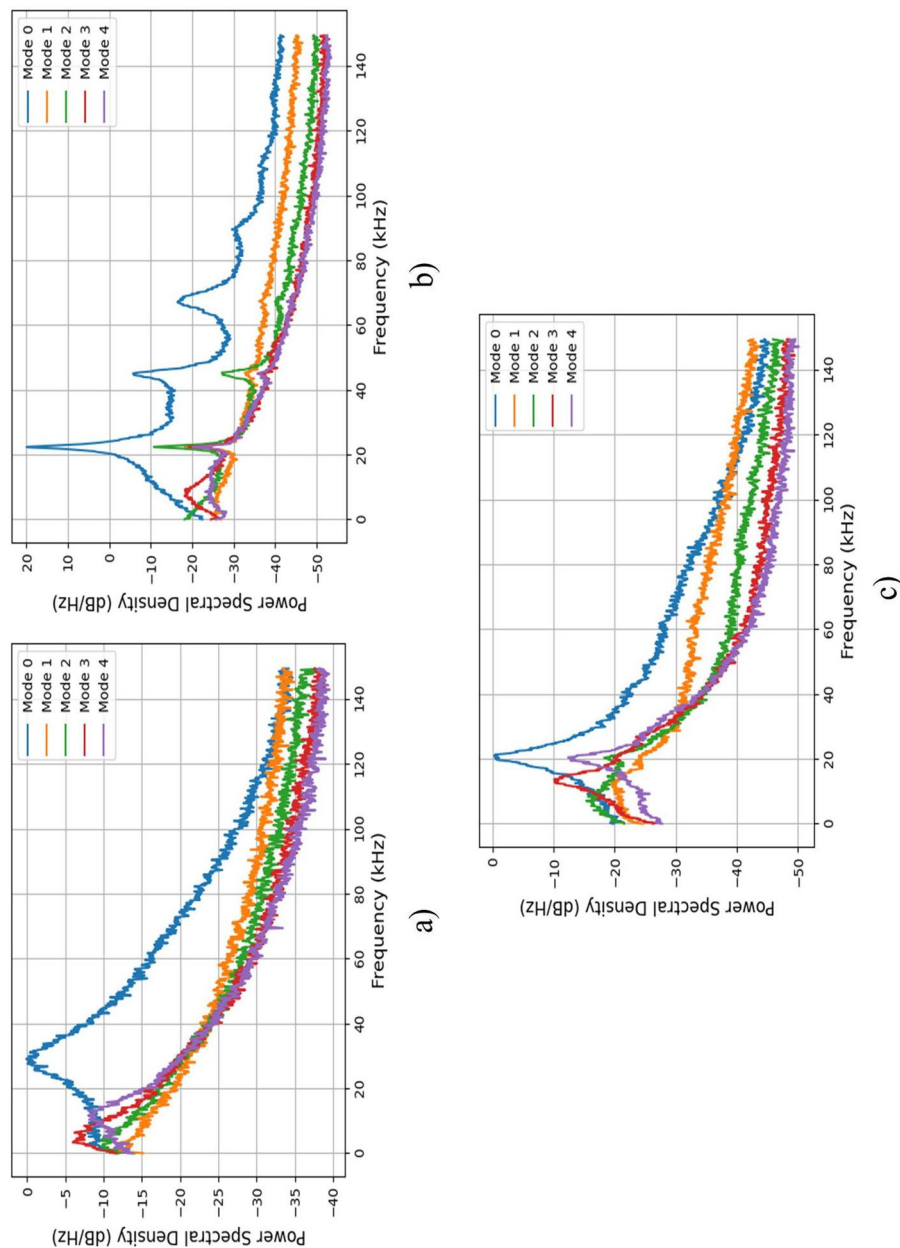


Fig. 3 Significant examples of the measured power spectral density corresponding to three different operating points: **a** $V_d = 300\text{ V}$, $I_d = 15\text{ A}$; **b** $V_d = 400\text{ V}$, $I_d = 7.5\text{ A}$; **c** $V_d = 400\text{ V}$, $I_d = 6.25\text{ A}$

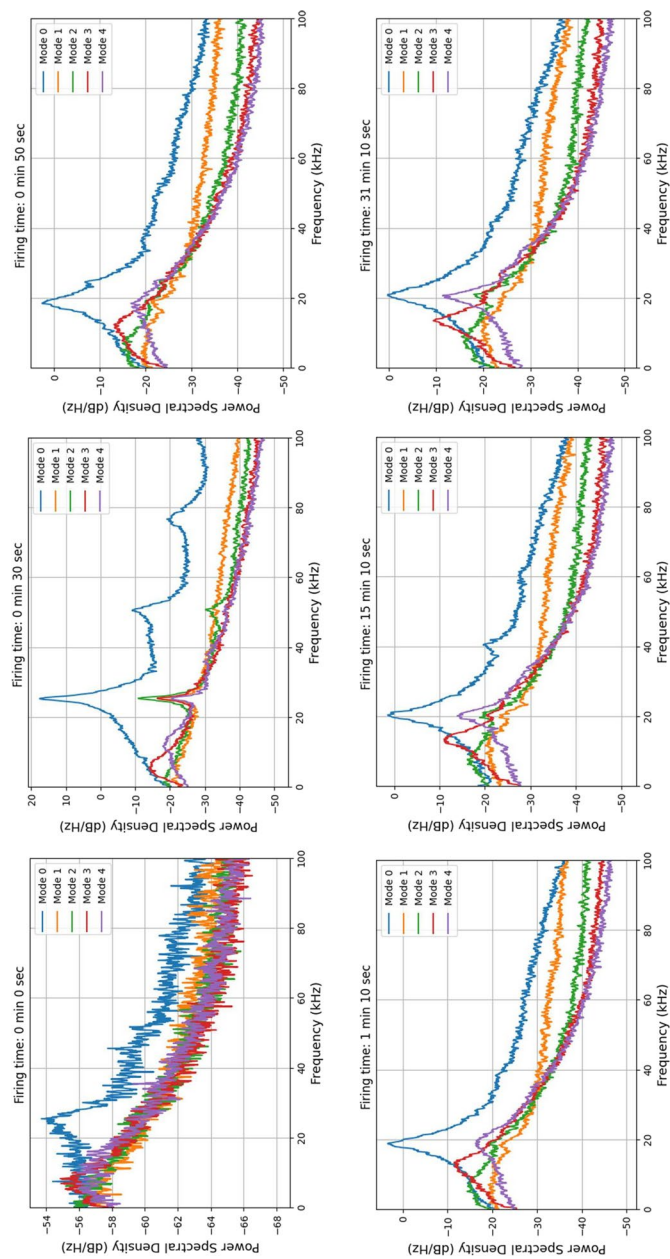


Fig. 4 Evolution of the breathing mode and the rotating azimuthal spokes mode from the first moments after the switching-on to the stationary condition

The series of measurements reported in Fig. 4 have been performed at high voltages, i.e. in the same regime where the presence of high order spokes modes is more probable. We report several power spectra, measured at varying time respect to the switching-on of the thruster. In the first 30 seconds of firing, the thruster exhibits a spectrum of the same type as the one reported in Fig. 3b, with a very sharp and strong breathing mode. Very shallow higher order modes are barely visible at lower frequencies. In the following minutes of firing, we observe a clear growing up of other azimuthal modes, in particular of order 3 and 4, whose peaks become sharper and higher. Simultaneously, the amplitude of the breathing oscillation drops by 2 order of magnitude and an increasing of the spoke frequencies is clearly visible.

The observed evolution of the rotational spectra can be interpreted as a redistribution of the oscillation energy among the different modes. In order to be more quantitative, the energy fraction associated to each mode has been calculated. In an oscillations at a specific angular velocity ω , the energy fraction is proportional to $A(\omega)\omega^2$, where $A(\omega)$ is the amplitude of the spectral density at such frequency. The numerical values of $A(\omega)$ can be extracted by fitting the spectrum with Lorentzian or Gaussian profiles. In Fig. 5a, we report the fraction of the instantaneous total energy associated to each mode as a function time. The reported values are estimated integrating the fitted spectra over the entire frequency windows and normalizing the sum to the unity at each instant. As can be seen, the energy reallocation is really clear. It exhibits an asymptotic behavior, where the changing is very drastic in the first seconds after the switching-on and then it seems to continue slower and slower also deeper in the stationary regime. Moreover, Fig. 5b confirms that there is also a continuous increasing of the mean spoke velocities, calculated as $v_m = 2\pi R f_m / m$, where f_m is the central frequency of the main peaks found in the spectrum, m is the mode order and R is the radius of the discharge channel.

The observed asymptotic behavior to a stationary condition is in agreement with previous parametric investigation of the rotating spokes properties performed by McDonald and Gallimore [11]. Their observations highlighted indeed that the thruster operates with the best performances when the spokes are present with frequencies smaller than the breathing modes. On the contrary, a condition where the breathing mode dominates does not corresponds to an efficient and stable working point. This different operating performances could explain why the observed spectra evolves from a dominating breathing-mode condition to a stationary regime where the presence of the spokes is stronger.

The understanding of the physical phenomena under the dynamical behavior of plasma rotating instabilities is out of the scope of this work but the reported results should be considered as an example of the potential perspective provided by the developed diagnostic system.

Conclusion

Azimuthal rotating instabilities that may produce turbulent and anomalous electron transport has been observed in various types of Hall thrusters, but a complete understanding of the phenomenon is far from being achieved.

In this paper, we proposed a diagnostic tool for the measurements of the characteristic frequencies of azimuthal instability. The proposed approach helps overcoming

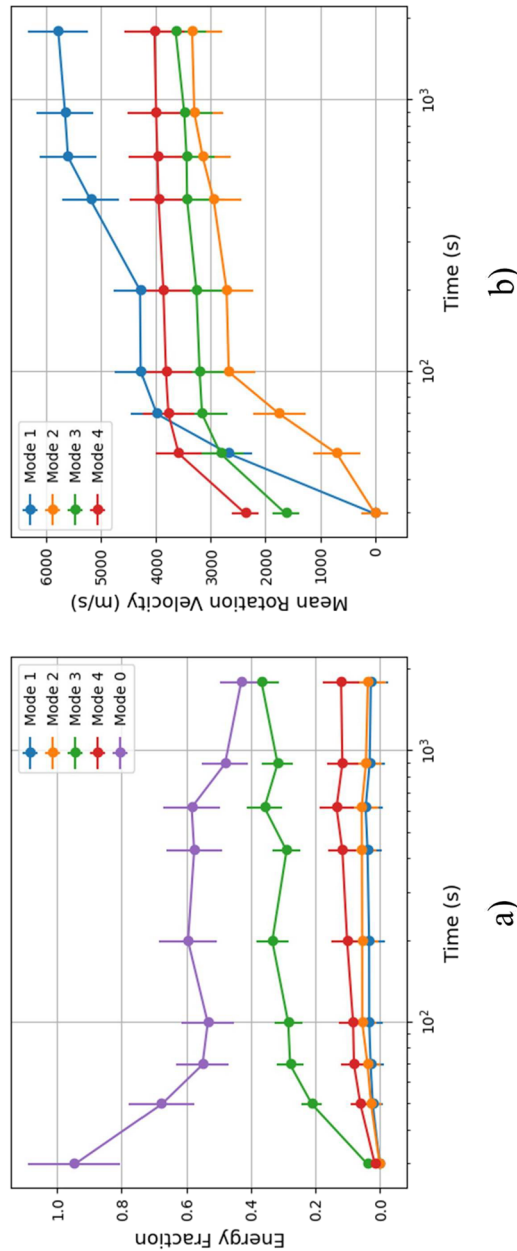


Fig. 5 Normalized fractional energy of the spokes modes, in (a), and rotational velocity of non-zero order modes, in (b), as a function of the time after the initial firing of the thruster (in logarithmic scale)

the limitations of the methods based on video acquisition by means of an ultra-fast camera, and it allows to reduce the requirements in terms of data flow-rate and computational efforts. The spatially resolved approach is anyway still preserved with a limited number of spatially resolved points.

The main advantage of the second approach is the possibility to develop a system that is able to measure the amplitude and the frequency of the rotating spokes instability in a real-time condition. In the current version, the reported diagnostics has a maximum bandwidth of the order of 400 kHz and a refresh time of the order of few seconds. Both of them can be improved quite easily changing the electronics, the digitizer and optimizing the computational algorithm.

In order to test the reliability and the performances of the diagnostics, several tests on HETs with a nominal power of 5 kW have been performed. The resulting Fourier analysis have shown a good signal-to-noise ratio and it has highlighted how different operating points can exhibit very different type of spectra. The relation between the observed type of spectra and the operating points seems to be in agreement with the previous experiments present in literature. In a last session of tests, we have also demonstrated how the real-time features of the developed diagnostic can help to understand more in detail the evolution of the energy and the velocities of the plasma rotating instabilities during the thruster operation.

Finally, the versatility of our setup makes it possible to adapt the setup to each specific measurement condition, opening the way to interesting perspectives. In particular, the use of dedicated sensors helps optimizing the sensitivity and facilitating the scalability of the whole set-up in order to acquire, for example, the light signal in different narrow spectral bands simultaneously or to develop diagnostic tools for direct-in-flight analysis.

Authors' contributions

L. M., M. P., A. M., Y. D., F. S. from Aerospazio Tecnologie contributed in the development of the diagnostic systems, the providing of the test facility and the data analysis. G. C., A. P. and V. V. from Safran contributed to the providing of the HET and its operation

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Data availability

Data are available upon reasonable request to the corresponding author.

Materials availability

Not applicable.

Code availability

Not applicable.

Declarations

Ethics approval and consent to participate

Not applicable.

Consent for publication

Not applicable.

Competing interests

The authors declare no competing interests.

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